



Next-Generation Biopesticides: Sustainable Innovations for Crop Protection

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Abstract

In recent years, major yield losses in agriculture are due to pests. In order to control them synthetic pesticides are used, but indiscriminate use leads to resistance and leads to risks to living organisms and the environment. Biopesticides, derived from microorganisms, plants, or minerals, are eco-friendly to the environment and provide sustainable results. This article reviews microbial, biochemical, and plant-incorporated biopesticides, along with innovative technologies such as RNAi, CRISPR, and nanotechnology. These innovations provide cost effective, eco-friendly, and sustainable crop protection.

Keywords: Biopesticides, RNAi, CRISPR-Cas, Nanoformulations, Microbial consortia, Plant Incorporated Protectants and Eco-friendly crop protection.

Introduction

Biopesticides are certain types of pesticides derived from such natural materials as animals, plants, bacteria and certain minerals (EPA, 2025). In other words, Biopesticides are living organisms, including parasites, predators, disease-causing fungi, bacteria and viruses, which are the natural enemies of the insect pests of the crops. Biopesticides are preferred because they leave little to no residue in the environment, are eco-friendly, do not harm non-target organisms such as pollinators and predators, are cost-effective, and help reduce the frequency of chemical pesticide applications. Global pesticide use increased between 2000 to 2023 by 71% to 3.7 million tons (FAO 2025). Pesticides include insecticides, fungicides, weedicides, disinfectants and substances used for preventing, destroying or controlling any pest harmful to human beings directly or indirectly. The pesticides are used to improve crop yield to ensure food security for the growing population. Every year, up to 40% of the yield is lost before they are harvested due to pests (CABI, 2025). Pesticides produce different types of toxicities, for instance, neurotoxicity, mutagenicity, carcinogenicity, teratogenicity, and endocrine disruption (Ahmad *et al.*, 2024). Pesticides also cause challenges due to resistance, toxicity, harm to beneficial organisms and leaving long-term residual effects in the environment. This article provides a comprehensive overview of biopesticides, discussing their classification, mechanisms of action, and emerging technologies that are reshaping crop protection.

Classes of biopesticides

There are three major types of biopesticides:

1) Microbial biopesticides

The successful application of *Bacillus thuringiensis* (Bt) and other microbial species has facilitated the identification of numerous new microbial taxa and strains, along with their bioactive toxins and virulence factors beneficial to the biopesticide sector. Key bacterial entomopathogens include species from the genera *Pseudomonas*, *Yersinia* and *Chromobacterium* (Chang *et al.*, 2003; Sporleder & Lacey, 2021). Fungal entomopathogens are represented by genera such as *Beauveria*, *Metarhizium*, *Verticillium*,

Lecanicillium, *Hirsutella*, and *Paecilomyces* (Chang *et al.*, 2003; Sporleder & Lacey, 2021). Additionally, entomopathogenic nematodes (EPNs) belonging to *Heterorhabditis* and *Steinernema* are vital biocontrol agents. They maintain symbiotic relationships with bacteria from the genera *Photorhabdus* and *Xenorhabdus* and are regarded as environmentally safe and non-toxic to mammals and non-target organisms (Chang *et al.*, 2003).

2) Biochemical biopesticides

Biochemical pesticides are naturally occurring substances that control pests by non-toxic mechanisms (EPA, 2025). Biochemical pesticides are classified into different types depending upon the functions in controlling insect pests by pheromones (semiochemicals), plant extracts and natural insect growth regulators (Kumar *et al.*, 2021). Pheromones affect the target pests olfactory system, thereby influencing mating, thus reducing the insect progeny. The essential oils from plants show promising insecticides and fungicidal properties due to the bioactive compounds (Bhavaniramya *et al.*, 2019). Active compounds from the plants can be used as phytopesticides including phenols, alkaloids and terpenes (Abubakar *et al.*, 2020). Insect growth regulators (IGRs) are insecticides that mimic hormones in young insects, thus disrupting insect growth and reproduction.

3) Plant Incorporated Protectants (PIPs)

Plant Incorporated Protectants are also known as Genetically Modified (GM) or Genetically Engineered (GE). PIPs are the pesticide substances that are produced by the plants, and the genetic materials are introduced from the foreign organisms to the plant (National Pesticide Information Centre, 2025). The typical example is the use of the Bt protein to develop PIPs through genetic engineering.

Emerging Technologies

RNAi-based Biopesticides

RNA interference (RNAi) or Post Transcriptional Gene Silencing (PTGS) is a conserved biological mechanism triggered by double-stranded RNA that suppresses the expression of specific genes. It serves as a Defence system against both internal parasitic elements and external pathogenic nucleic acids (Yadav *et al.*, 2025). The double-stranded RNA (dsRNA) based RNAi technology is becoming a revolutionary approach to deliver insect-resistant plant varieties as a Biopesticide. The dsRNA is produced in the Laboratory and is applied in the field through foliar spraying, root irrigation and trunk injection (He, Huang, & Tang, 2022). Topical application of dsRNA protects from many plant viruses across multiple families (Mitter *et al.*, 2017).

CRISPR-based Biopesticides

The CRISPR-Cas System operates through Cas enzymes such as Cas 9 or Cas 12 that act as molecular scissors, enabling targeted DNA modification. Gene drive Technology leverages CRISPR-Cas systems to spread targeted genetic modifications through pest populations, offering precise, eco-friendly pest control and resistance management (Komal *et al.*, 2023). CRISPR is applied to modify beneficial microorganisms, such as bacteria, fungi, for pest control. For instance, genetically engineered *Bacillus thuringiensis* (Bt) strains can be optimized to generate toxins, improving their lethality against specific pest species. CRISPR is used to overcome resistance to conventional pesticides by affecting target specific genes of pests. It is also used to engineer GM crops for pest control by releasing the repellent chemicals. In India mustard (*Brassica juncea*) seeds, CRISPR Cas 9-based editing provides less pungent and more resistant seeds to pests and diseases (Sankaranarayanan, 2024).

Nano Biopesticides

Biologically derived chemical compounds containing nanoparticles are used as pesticides. Nucleopolyhedrovirus (NPV), *Bacillus thuringiensis* (Bt), and entomopathogenic fungi (*Beauveria bassiana*, *Metarhiziumanisopliae*, etc.) are classical biopesticides. In nano-based biopesticides, these same agents can be combined with nanomaterials to improve stability, delivery, and efficacy (Mawcha *et al.*, 2024). The advantage of the Nano formulations is Controlled Release Formulation (CRF), which makes the precise and sustained delivery of bioactive compounds to control insect pests. The various types of nanoparticles include Metal nanoparticles such as silver (AgNps) and gold (AuNps), Inorganic Nanoparticles as silicon dioxide (SiO₂) Nps, and Nanopesticides based on plant extracts and essential oils. The Combination of Bt with modified SiO₂ nanoparticles increased mortality by 10-15% compared to Bt alone to control *Leptinotarsa decemlineata* and *Phyllotreta* sp., which demonstrates the potential of nanopesticides (Shatalova, Grizanova, & Dubovskiy, 2022).

Microbial Consortia

Microbial consortia include multiple microbial species with differing ecological preferences, such as soil type, host plant association, preferred colonization zones, and the specific pathogens they suppress (Dinis *et al.*, 2020). Even though individual microbes may act through distinct biological pathways, combining them in a consortium extends the range and intensity of pest and disease control (Kozieł *et al.*, 2023). The combination of these microorganisms promotes plant growth by overcoming various biotic stresses and thereby improves the efficacy. Even though the reach of microbial Consortia is limited due to its limited availability in the Market.

Future Prospects

In the future, biopesticides will play a crucial role in crop protection. Advanced molecular biology techniques, such as RNAi and CRISPR, will enable precise targeting of harmful pests without affecting beneficial organisms. Combining nanotechnology with biopesticides and microbial consortia provides effective results to control pests. Including the biopesticides with other pest protection management strategies will provide synergic results and reduce the dependency on synthetic pesticides. Various initiatives have to be taken to raise awareness among farmers about biopesticides. Thus, it biopesticides pave the way for overcoming pesticide resistance and its ill effects on human beings and the environment.

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